Network Protocol Analysis

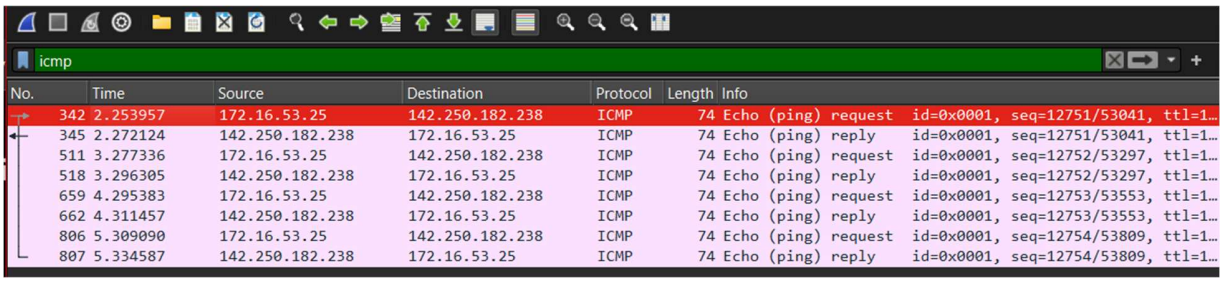
Assignment 2

Objective

To capture and analyze ICMP, TCP three-way handshake, UDP, DNS, HTTP, and ARP packets using Wireshark.

Output

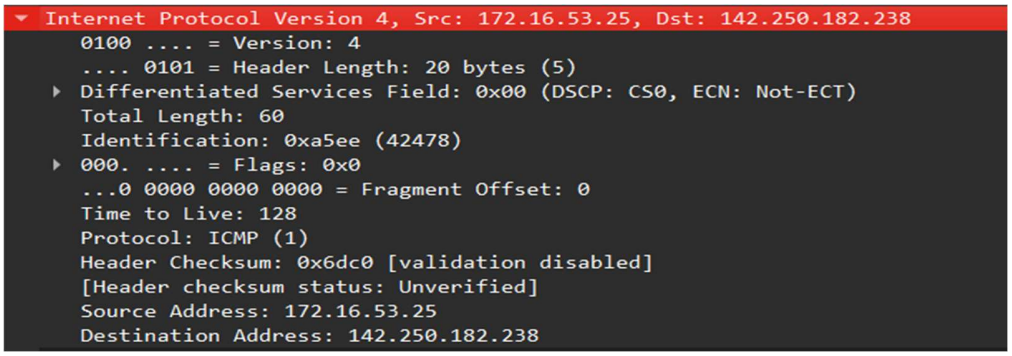
1: ICMP — Ping Request



icmp

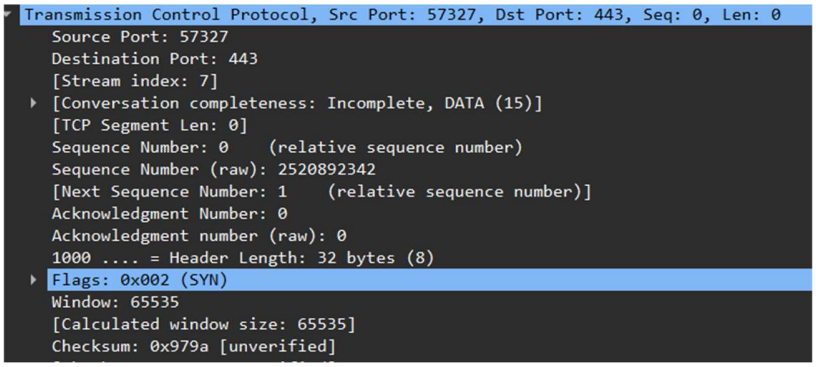
No.	Time	Source	Destination	Protocol	Length	Info
342	2.253957	172.16.53.25	142.250.182.238	ICMP	74	Echo (ping) request id=0x0001, seq=12751/53041, ttl=1…
345	2.272124	142.250.182.238	172.16.53.25	ICMP	74	Echo (ping) reply id=0x0001, seq=12751/53041, ttl=1…
511	3.277336	172.16.53.25	142.250.182.238	ICMP	74	Echo (ping) request id=0x0001, seq=12752/53297, ttl=1…
518	3.296305	142.250.182.238	172.16.53.25	ICMP	74	Echo (ping) reply id=0x0001, seq=12752/53297, ttl=1…
659	4.295383	172.16.53.25	142.250.182.238	ICMP	74	Echo (ping) request id=0x0001, seq=12753/53553, ttl=1…
662	4.311457	142.250.182.238	172.16.53.25	ICMP	74	Echo (ping) reply id=0x0001, seq=12753/53553, ttl=1…
806	5.309090	172.16.53.25	142.250.182.238	ICMP	74	Echo (ping) request id=0x0001, seq=12754/53809, ttl=1…
807	5.334587	142.250.182.238	172.16.53.25	ICMP	74	Echo (ping) reply id=0x0001, seq=12754/53809, ttl=1…

2: ICMP — Ping Reply



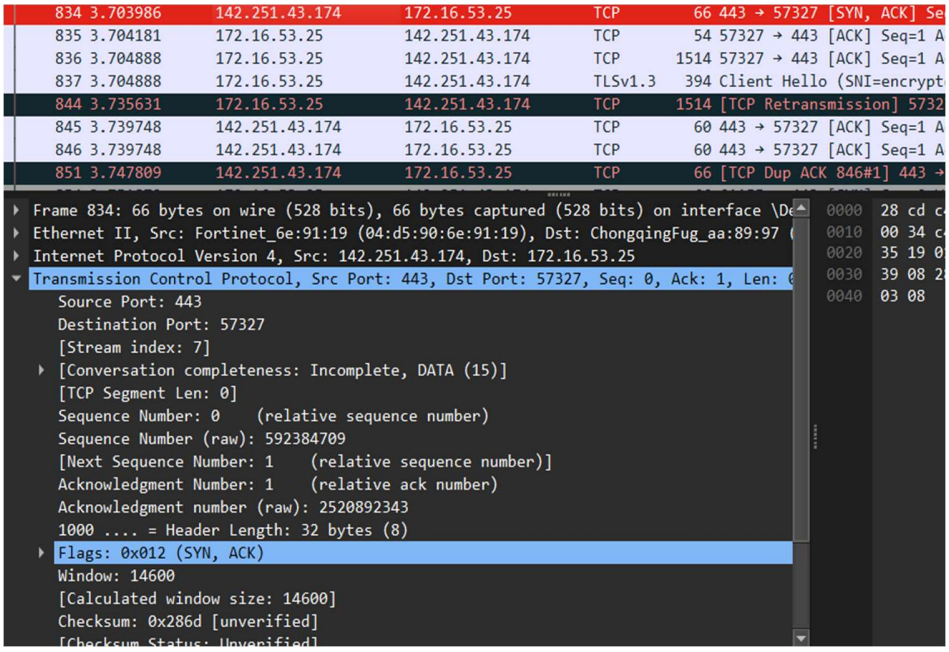
```
▾ Internet Protocol Version 4, Src: 172.16.53.25, Dst: 142.250.182.238
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 60
    Identification: 0xa5ee (42478)
  ▸ 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: ICMP (1)
    Header Checksum: 0x6dc0 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 172.16.53.25
    Destination Address: 142.250.182.238
```

3: TCP — SYN Packet

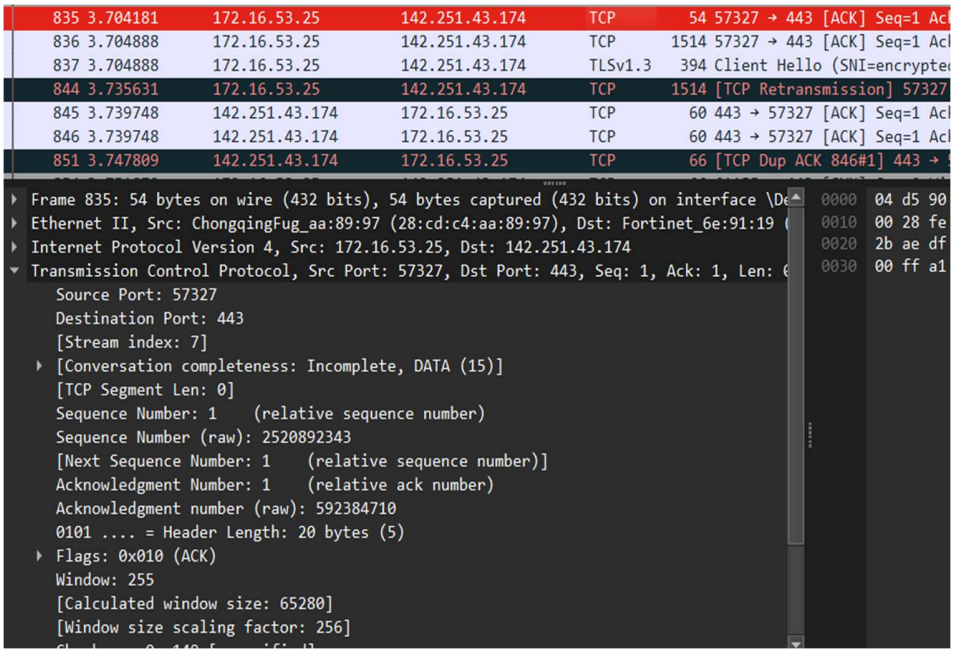


```
▾ Transmission Control Protocol, Src Port: 57327, Dst Port: 443, Seq: 0, Len: 0
    Source Port: 57327
    Destination Port: 443
    [Stream index: 7]
  ▸ [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 2520892342
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 0
    Acknowledgment number (raw): 0
    1000 .... = Header Length: 32 bytes (8)
  ▸ Flags: 0x002 (SYN)
    Window: 65535
    [Calculated window size: 65535]
    Checksum: 0x979a [unverified]
```

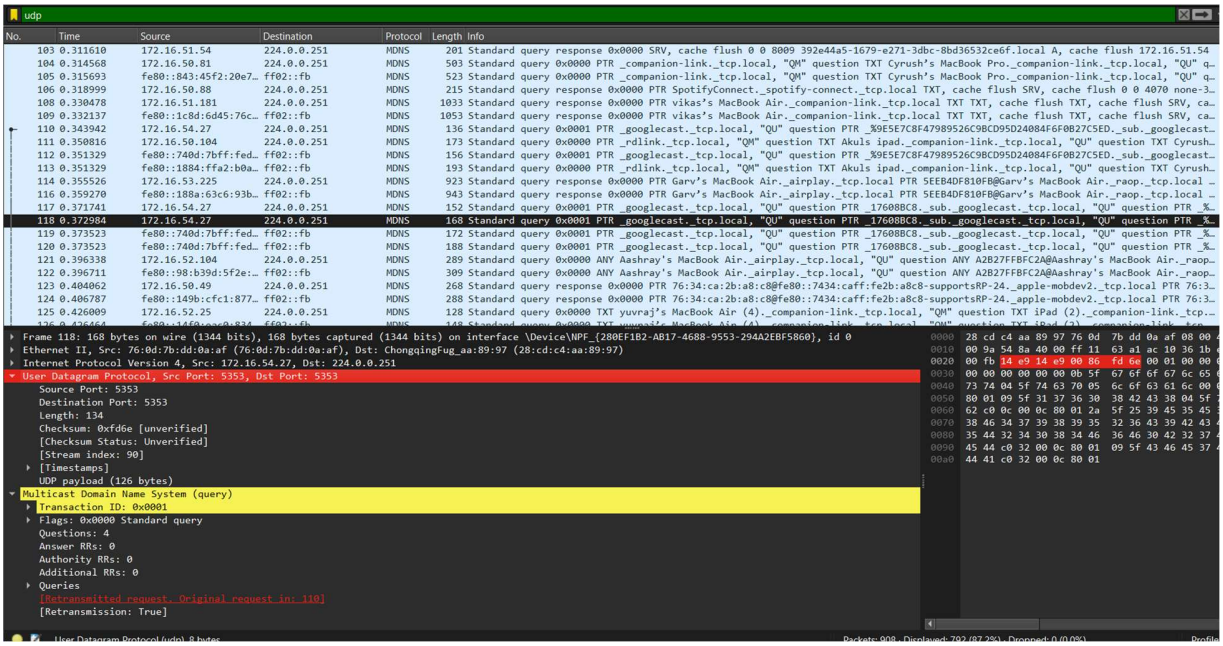
4: TCP — SYN-ACK Packet



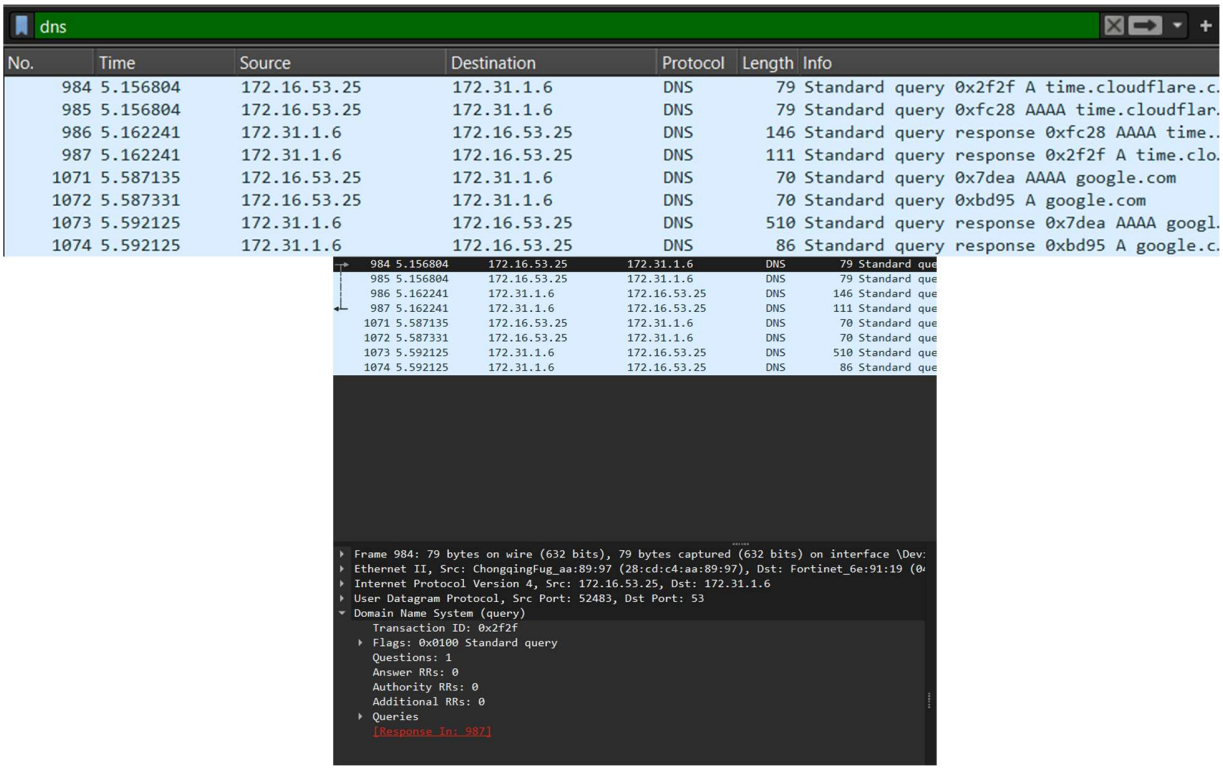
5: TCP — ACK Packet (Handshake Complete)



6: UDP Packet



7: DNS Query



8: DNS Response

No.	Time	Source	Destination	Protocol	Length	Info
986	5.162241	172.31.1.6	172.16.53.25	DNS	146	Standard query response 0x
987	5.162241	172.31.1.6	172.16.53.25	DNS	111	Standard query response 0x
1071	5.587135	172.16.53.25	172.31.1.6	DNS	70	Standard query 0x7dea AAAA
1072	5.587331	172.16.53.25	172.31.1.6	DNS	70	Standard query 0xbd95 A g
1073	5.592125	172.31.1.6	172.16.53.25	DNS	510	Standard query response 0x
1074	5.592125	172.31.1.6	172.16.53.25	DNS	86	Standard query response 0x

```
▸ Frame 986: 146 bytes on wire (1168 bits), 146 bytes captured (1168 bits) on interface '
▸ Ethernet II, Src: Fortinet_6e:91:19 (04:d5:90:6e:91:19), Dst: ChongqingFug_aa:89:97 (28
▸ Internet Protocol Version 4, Src: 172.31.1.6, Dst: 172.16.53.25
▸ User Datagram Protocol, Src Port: 53, Dst Port: 54991
▾ Domain Name System (response)
    Transaction ID: 0xfc28
  ▸ Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 2
    Authority RRs: 0
    Additional RRs: 1
  ▸ Queries
  ▸ Answers
  ▸ Additional records
    [Request In: 985]
    [Time: 0.005437000 seconds]

0000  28 cd c4
0010  00 84 72
0020  35 19 00
0030  00 02 00
0040  64 66 6c
0050  0c 00 1c
0060  f1 00 00
0070  01 00 00
0080  00 00 00
0090  00 00
```

9: HTTP Packets

http

No.	Time	Source	Destination	Protocol	Length	Info
342	1.265708	172.16.53.25	23.155.129.172	HTTP	581	GET /login/login.asp HTTP/1.1
467	1.570179	23.155.129.172	172.16.53.25	HTTP	732	HTTP/1.1 200 OK (text/html)

```
▸ Frame 342: 581 bytes on wire (4648 bits), 581 bytes captured (4648 bits) on interface \Device\NPF_{280EF1B2-AB17-4688-9553-294A2EBF5860}, id 0
▸ Ethernet II, Src: ChongqingFug_aa:89:97 (28:cd:c4:aa:89:97), Dst: Fortinet_6e:91:19 (04:d5:90:6e:91:19)
▸ Internet Protocol Version 4, Src: 172.16.53.25, Dst: 23.155.129.172
▸ Transmission Control Protocol, Src Port: 61396, Dst Port: 80, Seq: 1, Ack: 1, Len: 527
▾ Hypertext Transfer Protocol
  ▸ GET /login/login.asp HTTP/1.1\r\n
    Host: vbsca.ca\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
    Referer: https://www.google.com/\r\n
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9\r\n
  ▸ Cookie: ASPSESSIONIDACBQQACR=BBLKLFNAEKBELDAAFCIBJOCO\r\n
    \r\n
    [Full request URI: http://vbsca.ca/login/login.asp]
    [HTTP request 1/1]
    [Response in frame: 467]

0000  04 d5 90 6e 91 19 28 cd  c4 aa 89 97 08 00 45 00
0010  02 37 6a a1 40 00 80 06  13 af ac 10 35 19 17 9b
0020  81 ac ef d4 00 50 25 3e  2b 79 1d a8 3c 93 50 18
0030  00 ff f5 fb 00 00 47 45  54 20 2f 6c 6f 67 69 6e
0040  2f 6c 6f 67 69 6e 2e 61  73 70 20 48 54 54 50 2f
0050  31 2e 31 0d 0a 48 6f 73  74 3a 20 76 62 73 63 61
0060  2e 63 61 0d 0a 43 6f 6e  6e 65 63 74 69 6f 6e 3a
0070  20 6b 65 65 70 2d 61 6c  69 76 65 0d 0a 55 70 67
0080  72 61 64 65 2d 49 6e 73  65 63 75 72 65 2d 52 65
0090  71 75 65 73 74 73 3a 20  31 0d 0a 55 73 65 72 2d
00a0  41 67 65 6e 74 3a 20 4d  6f 7a 69 6c 6c 61 2f 35
00b0  2e 30 20 28 57 69 6e 64  6f 77 73 20 4e 54 20 31
00c0  30 2e 30 3b 20 57 69 6e  36 34 3b 20 78 36 34 29
00d0  20 41 70 70 6c 65 57 65  62 4b 69 74 2f 35 33 37
00e0  2e 33 36 20 28 4b 48 54  4d 4c 2c 20 6c 69 6b 65
00f0  20 47 65 63 6b 6f 29 20  43 68 72 6f 6d 65 2f 31
0100  34 37 2e 30 2e 30 2e 30  20 53 61 66 61 72 69 2f
0110  35 33 37 2e 33 36 0d 0a  41 63 63 65 70 74 3a 20
0120  74 65 78 74 2f 68 74 6d  6c 2c 61 70 70 6c 69 63
0130  61 74 69 6f 6e 2f 78 68  74 6d 6c 2b 78 6d 6c 2c
```

10: ARP Request & Reply

arp

No.	Time	Source	Destination	Protocol	Length	Info
91	1.532646	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
96	2.292872	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
102	3.282048	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
103	4.298448	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
104	5.286776	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
107	6.290745	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
112	7.298439	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
134	8.283832	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
136	9.289302	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
142	10.298956	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
146	11.293560	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131
147	12.285999	ChongqingFug_aa:89:…	Broadcast	ARP	42	Who has 10.50.73.130? Tell 10.50.73.131

```
▸ Frame 91: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{280EF1B2-AB17-4688-9553-294A2EBF5860}, id 0
▸ Ethernet II, Src: ChongqingFug_aa:89:97 (28:cd:c4:aa:89:97), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▾ Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: ChongqingFug_aa:89:97 (28:cd:c4:aa:89:97)
    Sender IP address: 10.50.73.131
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 10.50.73.130
```

11: Source & Destination MAC/IP Summary

317	2.256984	148.113.16.46	10.50.73.131	TCP	66	443 → 50262 [ACK] Seq=1 Ack=2 Win=501 Len=0 SLE=1 SRE=2
318	2.256984	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
319	2.256984	142.251.157.119	10.50.73.131	UDP	137	443 → 61979 Len=95
320	2.256984	64:ff9b::8efb:dd62	2409:40d1:0:9dc3:4d…	QUIC	87	Protected Payload (KP0)
321	2.256984	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
322	2.256984	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
323	2.257241	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
324	2.258892	10.50.73.131	142.251.157.119	UDP	86	61979 → 443 Len=44
325	2.259032	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
326	2.271864	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
327	2.271864	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
328	2.271864	142.251.157.119	10.50.73.131	UDP	309	443 → 61979 Len=267

```
▸ Frame 317: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{280EF1B2-AB17-4688-9553-294A2EBF5860}, id 0
▾ Ethernet II, Src: 72:9c:93:3c:8c:f4 (72:9c:93:3c:8c:f4), Dst: ChongqingFug_aa:89:97 (28:cd:c4:aa:89:97)
  ▸ Destination: ChongqingFug_aa:89:97 (28:cd:c4:aa:89:97)
  ▸ Source: 72:9c:93:3c:8c:f4 (72:9c:93:3c:8c:f4)
    Type: IPv4 (0x0800)
▸ Internet Protocol Version 4, Src: 148.113.16.46, Dst: 10.50.73.131
▸ Transmission Control Protocol, Src Port: 443, Dst Port: 50262, Seq: 1, Ack: 2, Len: 0
```

317	2.256984	148.113.16.46	10.50.73.131	TCP	66	443 → 50262 [ACK] Seq=1 Ack=2 Win=501 Len=0 SLE=1 SRE=2
318	2.256984	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
319	2.256984	142.251.157.119	10.50.73.131	UDP	137	443 → 61979 Len=95
320	2.256984	64:ff9b::8efb:dd62	2409:40d1:0:9dc3:4d…	QUIC	87	Protected Payload (KP0)
321	2.256984	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
322	2.256984	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
323	2.257241	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
324	2.258892	10.50.73.131	142.251.157.119	UDP	86	61979 → 443 Len=44
325	2.259032	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
326	2.271864	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
327	2.271864	142.251.157.119	10.50.73.131	UDP	1292	443 → 61979 Len=1250
328	2.271864	142.251.157.119	10.50.73.131	UDP	309	443 → 61979 Len=267

```
▸ Frame 317: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{280EF1B2-AB17-4688-9553-294A2EBF5860}, id 0
▾ Ethernet II, Src: 72:9c:93:3c:8c:f4 (72:9c:93:3c:8c:f4), Dst: ChongqingFug_aa:89:97 (28:cd:c4:aa:89:97)
  ▸ Destination: ChongqingFug_aa:89:97 (28:cd:c4:aa:89:97)
  ▸ Source: 72:9c:93:3c:8c:f4 (72:9c:93:3c:8c:f4)
    Type: IPv4 (0x0800)
▾ Internet Protocol Version 4, Src: 148.113.16.46, Dst: 10.50.73.131
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 52
    Identification: 0x0000 (0)
  ▸ 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 43
    Protocol: TCP (6)
    Header Checksum: 0x5770 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 148.113.16.46
    Destination Address: 10.50.73.131
▸ Transmission Control Protocol, Src Port: 443, Dst Port: 50262, Seq: 1, Ack: 2, Len: 0
```

```
0000  28 cd
0010  00 34
0020  49 83
0030  01 f5
0040  78 4a
```